

CP2.5 — Polar Coordinates

1. Introduction to Polar Coordinates

A point P has **polar coordinates** (r, θ) where:

- r = distance from origin (pole)
- θ = angle measured anticlockwise from initial line (positive x-axis)

Convention: $r \geq 0$, $-\pi < \theta \leq \pi$ (or $0 \leq \theta < 2\pi$).

1.1 Converting Between Polar and Cartesian

$$x = r \cos \theta, \quad y = r \sin \theta, \quad r^2 = x^2 + y^2, \quad \tan \theta = \frac{y}{x}$$

Example: Convert $(3, \pi/3)$ to Cartesian.

$$x = 3 \cos(\pi/3) = 3/2, \quad y = 3 \sin(\pi/3) = 3\sqrt{3}/2.$$

Example: Convert $(-1, \sqrt{3})$ to polar.

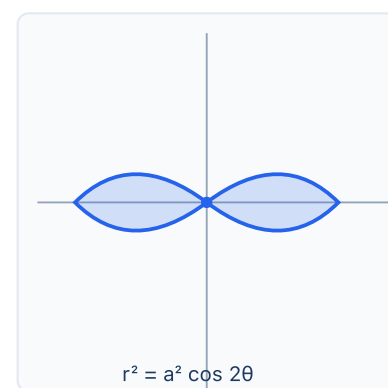
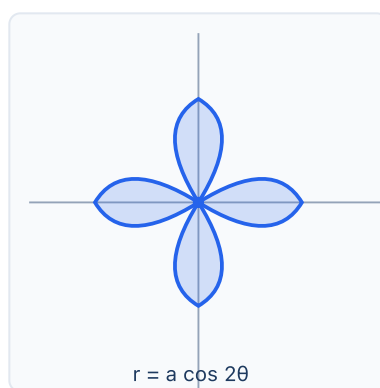
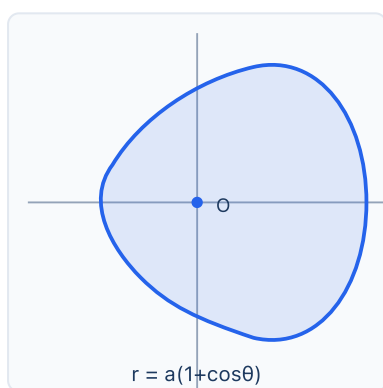
$$r = \sqrt{1+3} = 2, \quad \theta = \arctan(\sqrt{3}/(-1)). \text{ Point is in Q2, so } \theta = \pi - \pi/3 = 2\pi/3.$$

2. Sketching Polar Curves

2.1 Key Polar Curves — Summary Table

Equation	Curve	Key features
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$r = a$	Circle	Centre at origin, radius a
$r = 2a \cos \theta$	Circle	Centre $(a, 0)$, radius a ; passes through origin
$r = 2a \sin \theta$	Circle	Centre $(0, a)$, radius a ; passes through origin
$r = a(1 + \cos \theta)$	Cardioid	Cusp at origin, max $r = 2a$ at $\theta = 0$
$r = a \cos n\theta$	Rose (n petals if odd, $2n$ if even)	Each petal length a
$r = a\theta$	Archimedean spiral	Coils outward from origin
$r^2 = a^2 \cos 2\theta$	Lemniscate	Figure-of-eight; exists only for $ \theta \leq \pi/4$



2.2 Sketching Technique

1. Make a table of θ values (e.g., $0, \pi/6, \pi/4, \pi/3, \pi/2, \pi, \dots$) and compute r
2. Note symmetry: if $r(\theta) = r(-\theta)$, symmetric about initial line; if $r(\theta) = r(\pi - \theta)$, symmetric about $\theta = \pi/2$
3. Note where $r = 0$ (curve passes through origin)
4. Note where r is maximum
5. Plot and connect smoothly

Exam tip: For rose curves $r = a \cos n\theta$: the curve has n petals if n is odd, and $2n$ petals if n is even. Each petal has maximum length a .

3. Area Enclosed by a Polar Curve

The area enclosed by $r = f(\theta)$ between $\theta = \alpha$ and $\theta = \beta$ is:

$$A = \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta$$

Derivation: A small sector with angle $d\theta$ and radius r has area $\approx \frac{1}{2}r^2 d\theta$. Summing gives the integral.

Example: Find the area enclosed by the cardioid $r = a(1 + \cos \theta)$.

By symmetry about $\theta = 0$:

$$A = 2 \cdot \frac{1}{2} \int_0^{\pi} a^2(1 + \cos \theta)^2 d\theta = a^2 \int_0^{\pi} (1 + 2 \cos \theta + \cos^2 \theta) d\theta$$

Using $\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$:

$$\begin{aligned} &= a^2 \int_0^{\pi} \left(\frac{3}{2} + 2 \cos \theta + \frac{\cos 2\theta}{2} \right) d\theta \\ &= a^2 \left[\frac{3\theta}{2} + 2 \sin \theta + \frac{\sin 2\theta}{4} \right]_0^{\pi} = a^2 \left(\frac{3\pi}{2} \right) = \frac{3\pi a^2}{2} \end{aligned}$$

Example: Find the area of one petal of $r = a \cos 2\theta$.

One petal in $-\pi/4 \leq \theta \leq \pi/4$ (where $\cos 2\theta \geq 0$):

$$\begin{aligned} A &= \frac{1}{2} \int_{-\pi/4}^{\pi/4} a^2 \cos^2 2\theta d\theta = a^2 \int_0^{\pi/4} \cos^2 2\theta d\theta \\ &= a^2 \int_0^{\pi/4} \frac{1 + \cos 4\theta}{2} d\theta = a^2 \left[\frac{\theta}{2} + \frac{\sin 4\theta}{8} \right]_0^{\pi/4} = a^2 \cdot \frac{\pi}{8} = \frac{\pi a^2}{8} \end{aligned}$$

4. Area Between Two Polar Curves

The area between $r = f(\theta)$ (outer) and $r = g(\theta)$ (inner) for $\alpha \leq \theta \leq \beta$:

$$A = \frac{1}{2} \int_{\alpha}^{\beta} [f(\theta)^2 - g(\theta)^2] d\theta = \frac{1}{2} \int_{\alpha}^{\beta} (r_1^2 - r_2^2) d\theta$$

Example: Find the area inside $r = 2 \cos \theta$ but outside $r = 1$.

Curves intersect where $2 \cos \theta = 1$, i.e. $\theta = \pm\pi/3$.

By symmetry:

$$\begin{aligned} A &= 2 \cdot \frac{1}{2} \int_0^{\pi/3} [(2 \cos \theta)^2 - 1^2] d\theta = \int_0^{\pi/3} (4 \cos^2 \theta - 1) d\theta \\ &= \int_0^{\pi/3} (2 + 2 \cos 2\theta - 1) d\theta = \int_0^{\pi/3} (1 + 2 \cos 2\theta) d\theta \\ &= [\theta + \sin 2\theta]_0^{\pi/3} = \frac{\pi}{3} + \frac{\sqrt{3}}{2} \end{aligned}$$

Exam tip: Always find intersection points first by solving $r_1 = r_2$ (and check $r = 0$ cases). Sketch the curves to identify which is outer/inner and the correct range of θ .

5. Converting Polar Equations to Cartesian

Example: Show that $r = 2a \cos \theta$ is a circle in Cartesian form.

Multiply both sides by r : $r^2 = 2ar \cos \theta$.

Substitute: $x^2 + y^2 = 2ax$, i.e. $(x - a)^2 + y^2 = a^2$. Circle, centre $(a, 0)$, radius a . ✓

Example: Convert $r = \frac{3}{2 - \cos \theta}$ to Cartesian.

$$r(2 - \cos \theta) = 3 \Rightarrow 2r - r \cos \theta = 3 \Rightarrow 2\sqrt{x^2 + y^2} - x = 3$$

$$2\sqrt{x^2 + y^2} = x + 3 \Rightarrow 4(x^2 + y^2) = (x + 3)^2$$

6. Tangents to Polar Curves

Convert to parametric with θ as parameter: $x = r \cos \theta$, $y = r \sin \theta$.

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$$

where:

$$\frac{dy}{d\theta} = \frac{dr}{d\theta} \sin \theta + r \cos \theta, \quad \frac{dx}{d\theta} = \frac{dr}{d\theta} \cos \theta - r \sin \theta$$

Tangents parallel to initial line ($y = 0$ direction, i.e. horizontal): set $\frac{dy}{d\theta} = 0$ and $\frac{dx}{d\theta} \neq 0$.

Tangents perpendicular to initial line (vertical): set $\frac{dx}{d\theta} = 0$ and $\frac{dy}{d\theta} \neq 0$.

Example: For the cardioid $r = a(1 + \cos \theta)$, find $\frac{dy}{dx}$ and the angles at which tangents are parallel to the initial line.

$$\frac{dr}{d\theta} = -a \sin \theta.$$

$$\frac{dy}{d\theta} = -a \sin \theta \cdot \sin \theta + a(1 + \cos \theta) \cos \theta = a(\cos \theta + \cos 2\theta)$$

$$\text{Set } \frac{dy}{d\theta} = 0: \cos \theta + \cos 2\theta = 0, \text{ i.e. } \cos \theta + 2\cos^2 \theta - 1 = 0.$$

$$\text{Let } c = \cos \theta: 2c^2 + c - 1 = 0 \Rightarrow (2c - 1)(c + 1) = 0.$$

$$c = 1/2 \ (\theta = \pi/3, 5\pi/3) \text{ or } c = -1 \ (\theta = \pi, \text{ but check } dx/d\theta \text{ at this point}).$$

7. Exam-Style Practice Questions

1. **[Conversion]** Convert the polar equation $r = 4 \sin \theta$ to Cartesian form and describe the curve.
2. **[Area]** Find the total area enclosed by the curve $r = 2 + 3 \sin \theta$ between $\theta = 0$ and $\theta = 2\pi$.
3. **[Area — rose]** Find the total area enclosed by all the petals of $r = 4 \cos 3\theta$.
4. **[Area between curves]** Find the area of the region that is inside $r = 3 \cos \theta$ and outside $r = 1 + \cos \theta$.
5. **[Tangent]** For the curve $r = 1 - \cos \theta$, find the Cartesian equation of the tangent at $\theta = \pi/2$.
6. **[Sketch]** Sketch the curve $r = 2 + \cos \theta$, labelling key features. State the range of r .

General exam tips for CP2.5:

- Area formula: $\frac{1}{2} \int r^2 d\theta$. The $\frac{1}{2}$ is frequently forgotten — costs all subsequent marks.
- Use symmetry where possible — halve the interval and double — saves integration work.
- For tangent questions: differentiate $y = r \sin \theta$ and $x = r \cos \theta$ with respect to θ using product rule.
- Lemniscate: exists only for $\cos 2\theta \geq 0$, i.e. only certain θ ranges give real r .